

Emergency Hospital Admission Prediction - Written Responses

Problem Definition

Emergency department (ED) overcrowding has become a significant healthcare challenge across Canada and particularly in Atlantic provinces such as New Brunswick and Nova Scotia. Hospitals in the region face increasing patient volumes, long wait times, and limited clinical resources. For example, Canada has some of the longest emergency wait times among comparable countries, with 29% of patients reporting waits of more than four hours in emergency departments. In New Brunswick specifically, an analysis of nearly 1.5 million emergency visits between 2020 and 2024 found that about 66% of patients were not seen within the recommended triage timelines, highlighting systemic pressure on emergency services. ER departments become overcrowded when demand for urgent care exceeds available hospital capacity.

I built a classification model to predict whether a hospital admission is likely to be an emergency case using patient-level data and machine learning techniques. A successful outcome would be a model that achieves an F1 score above 0.50–0.60 with high recall for emergency cases ($\geq 70\%$), ensuring that most true emergency admissions are correctly identified. Additionally, a ROC-AUC closer to 0.70–0.80 would indicate a good ability to distinguish between emergency and non-emergency cases. While an F1 score above 0.80 and AUC close to 1.0 would represent excellent performance, such results typically require richer clinical data.

Approach & Tools

I approached this problem as a binary classification task to predict whether a hospital admission is likely to be an emergency case. I first conducted exploratory data analysis and engineered the target variable (`EmergencyFlag`) from the existing *Admission Type* variable, converting it into a binary outcome suitable for classification modeling. Using Python, Pandas, and Scikit-learn, I built preprocessing pipelines with `StandardScaler`, `OneHotEncoder`, and `OrdinalEncoder`, and addressed class imbalance using SMOTE. I evaluated several models including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost, comparing their performance using F1-score and

recall, along with cross-validation to ensure model robustness. I selected these approaches because classification models are well suited for predicting binary healthcare outcomes, pipelines ensure consistent preprocessing during model training and validation while performance metric selected to keep false negative as low as possible. I also considered alternative approaches such as time-series forecasting of emergency department demand or more advanced ensemble models, but focused on classification because the goal was to predict the likelihood of emergency admission at the patient level.

Reflection

Understanding and preparing the dataset accounted for nearly 60% of the project time, far more than initially anticipated. My original plan was to use the Canadian Community Health Survey (CCHS) data from Statistics Canada; however, I encountered significant challenges as the data dictionary did not clearly align with variable names, requiring me to manually interpret and map over 200 variables, which was both time-consuming and error-prone. Due to these constraints, I transitioned to a synthetic Kaggle dataset that was easier to work with but lacked key clinical features that truly drive emergency department (ED) admissions. This limitation was reflected in the model performance, with the best model achieving an F1 score of approximately 0.48. This experience reinforced that data quality and relevance are more critical than model complexity, as even well-tuned models cannot perform well without meaningful predictors. What worked well was the development of a structured machine learning pipeline and systematic comparison of multiple models. In future work, I would prioritize early validation of data sources, clearer alignment with domain-specific variables, and stakeholder engagement to identify key predictive features, rather than focusing heavily on model tuning at later stages.